University of Essex
Department of Economics
Dr. Neslihan Sakarya

Session 2024-2025
Autumn Term

EC466: Estimation and Inference in Econometrics

Worksheet # 1

Let $\mathbf{y}$ be an $n \times 1$ vector, $\mathbf{X}$ be an $n \times K$ matrix, β is a $K \times 1$ vector of parameters, and ε is $n \times 1$ vector of errors. We write a regression equation in the vector form as follows

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon.$$

1. Write down the linear regression equation for $\mathbf{y}'$.

2. What are the numbers of rows and columns of $\mathbf{X}'\mathbf{y}$?

3. Let $\mathbf{X} = [\mathbf{x_1}, \mathbf{x_2}, \ldots, \mathbf{x_n}]'$ where $\mathbf{x_i}$ is a $K \times 1$ vector and the i^{th} row of $\mathbf{X}$. Write down $\mathbf{X}'\mathbf{y}$ in terms of $\mathbf{x_i}$ and y_i where y_i is the i^{th} element of $\mathbf{y}$.

4. Is $\mathbf{X}'\mathbf{y}$ invertible?

5. What are the numbers of rows and columns of $\mathbf{X}'\mathbf{X}$?

6. What are the numbers of rows and columns of $\mathbf{X}\mathbf{X}'$?

7. Suppose that $\mathbf{X}$ is full column rank.

   (a) What is the rank of $\mathbf{X}'\mathbf{X}$?

   (b) What is the rank of $\mathbf{X}\mathbf{X}'$?

   (c) Is $\mathbf{X}'\mathbf{X}$ invertible?

   (d) Is $\mathbf{X}\mathbf{X}'$ invertible?